Osmo Gas Dryer V2 Productization Proposal

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Introduction

Osmo's Ion Mobility Spectrometry based sensor requires zero air. This requires air with near zero moisture or hydrocarbons. Osmo has a functional model that allows for this level of filtering, however it is unsuitable for use in many environments. This design is intended to utilize a dual-column desiccant swing architecture to create a more versatile model that meets the following requirements:

- Output consistently reaches <1ppm water and <0.1ppm total hydrocarbons
- Output operates at 800mbar pressure
- System must fit within a 11”x13”x5.5” enclosure
- System must be fully autonomous for at least 1 month
- Integrated safety features for power interruptions and thermal runaway

Pneumatic Architecture

The pneumatic architecture used for this design is similar to that of current dual-column desiccant swing models, with a few notable additions. The system utilizes a shuttle valve to restrict flow to one desiccant column at a time and uses solenoid valves to control the direction of flow in each branch of the system. Rough drawings of the system as well as the pneumatic logic are shown in Figures 1 and 2.

Flow is pushed into the system by a compressor where it enters a shuttle valve. Solenoid valves on each branch with opposite states ensure pressure differentials guide the shuttle valve to its correct position. Air is then pushed through the desiccant tower where it exits to another shuttle valve. This shuttle valve is similarly controlled by pressure gradients due to the solenoid valves. A small portion of air from the exit is passed to the other desiccant branch to assist with desorption.

Pressure from the compressor is sufficiently high that even with head loss due to pipes and the desiccant, flow will still exceed the 800mbar requirement at the exit. To ensure the flow meets the required pressure exactly, a bleed valve is installed to reduce pressure before exiting the system.

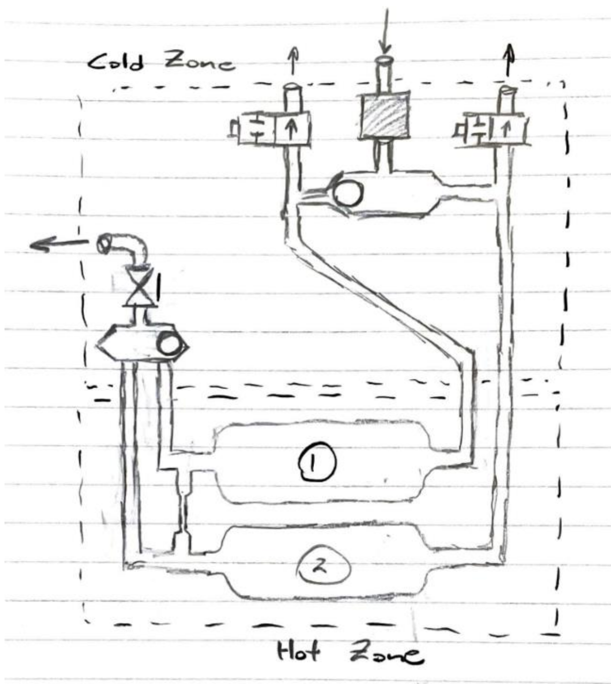

Figure 1 – Sketch of System with Defined Hot and Cold Zones

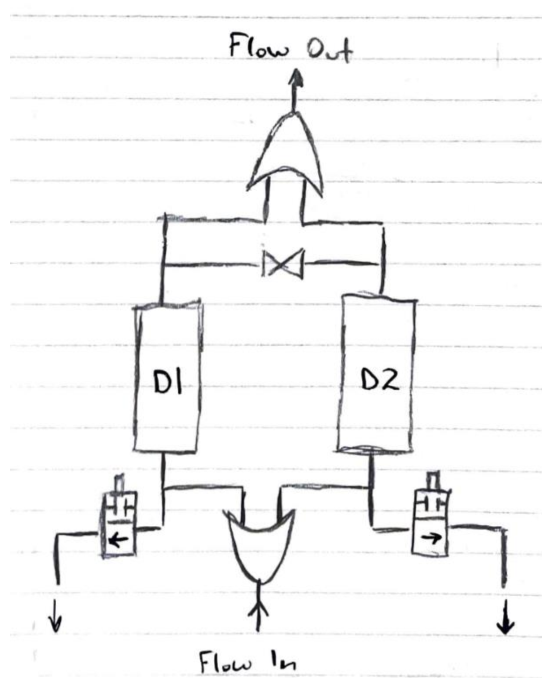

Figure 2 – Pneumatic Logic Diagram

Mechatronic Architecture

The system is controlled by an ESP32 microcontroller. This board was chosen due to its extra features and more powerful processing, as well as my familiarity with it from previous similar projects. A block diagram of the control logic is shown in Figure 3.

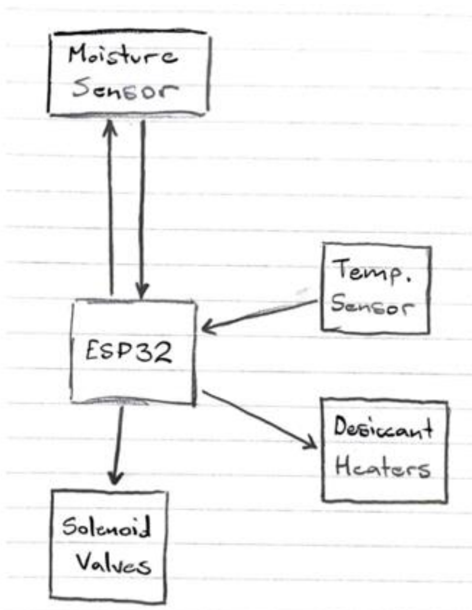

Figure 3 – Block Diagram of Control Logic

The ESP32 receives data from the moisture sensor placed at the exit of the system as well as the thermocouples placed along the length of the desiccant columns.

Data from the moisture sensor is used to trigger the swing between desiccants. When the reading reaches a dew point value nearly corresponding to the limit ppm, a relay is triggered, switching the position of both solenoids.

A second relay is also triggered, sending current through the silicon rubber flexible electric-resistive heaters attached to the desiccant in its desorption phase. Thermocouples are then monitored to ensure proper heating is occurring.

The swing logic is not timed and instead relies on moisture data to trigger swings, avoiding hysteresis in the system.

Desiccant Design

The limiting factor for the desiccant design was the volume it could occupy. To meet the filtering requirements, a layered molecular sieve design was used in place of the singular desiccant design. These ordered layers consisted of two layers of 13X Molecular Sieve, one layer of 4A Molecular Sieve, and one layer of 3A Molecular Sieve. These layers met the requirements of the system while occupying a significantly smaller space than typical desiccant materials. A rough sketch of the desiccant is shown in Figure 4.

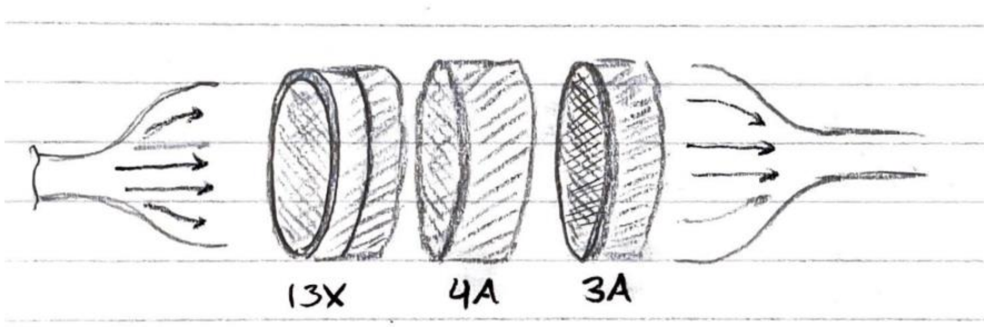

Figure 4 – Molecular Sieve Ordered Layers

13X was selected due to its high-capacity water absorption ability, as well as its fast desorption. Once majority of the water had been removed, the 4A layer removed further water as well as hydrocarbons. The 3A layer is ideal for very low ppm air and brings air to required conditions. All of these filters are able to undergo desorption at 250°C.

These layers are encased in aluminum, allowing for rapid heat conduction. There is also a fine steel mesh containing the faces of the layers while allowing for a small margin of expansion during heating.

The volume required for each layer was calculated using the total water load and working absorption capacity of each layer. From this, the desiccant structure would need to be a minimum of 3.425in in length assuming a ~50in^2 surface area. At these sizes, the time for desorption was estimated to be 50-80 minutes, depending on the safety factor applied.

Power and Thermal Management

To protect from high thermal loads, the system is divided into a "Hot Zone" and "Cold Zone", as shown in Figure 1. The Hot Zone is insulated with a ceramic barrier, and case fans are installed to force convective cooling within the system.

Power consumption of the system was kept within a range achievable in most environments. A summary of power usage is shown below in Table 1.

Table 1 – Power Usage By Component

Name	Voltage (V)	Current (A)	Quantity	Total Power (W)
Compressor	24	2	1	48
ESP32	5	0.25	1	1.25
Heating	-	-	2	~600 W
Solenoids	8	2	2	32
Case Fans	12	0.001	4	0.048

Component Selection

An itemized list of components with accompanying descriptions is shown below in Table 2.

Name	Quantity	Description
P05H012A-BLDC-C	1	Compressor used to drive system
Solenoid Valve	2	Connected to ESP32 to allow for flow control
Check Valve	2	Downstream of Solenoid valves, prevent backflow
Bleed Valve	1	Ensure pressure is at 800mbar at exit
Shuttle Valve	2	Provide Or Gate logic to pipe system
Silicon Flexible Resistive Heaters	6	Heat desiccant layers during down swing
Thermocouples	6	Give temperature data to ESP32, avoid thermal runaway
Case Fans	2	Provide forced convective cooling to system
PhyMetrix PLMa Loop Powered Dew Point Transmitter	1	Moisture sensor to monitor outlet conditions and trigger desiccant swing

Reliability Logic

The system is equipped to handle power failures and thermal runaway.

The ESP32 is equipped with non-volatile memory, allowing for the storing of data even when power is lost. The swing architecture of this system is controlled by a downstream moisture sensor, so only one variable needs to be stored to check the previous state of the system. If power failure were to occur, the system would not act any differently once power had been regained.

Thermal runaway is monitored by multiple thermocouples in the system. If the PID feedback loop detects overheating, the heating systems will be shut down until the fans cool it to the proper temperature and flow will be directed away from the sensor.